

| Question |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                                         | Marks available |     |     |       |       |      |
|----------|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                                         | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 1        | (a) |       | Ductile: Can be drawn into wires <b>or</b> can be easily deformed [plastically] (1) Don't accept reference to malleability<br>Steel: addition of [carbon] atoms inhibits movement / correct reference to impact of atoms on movement (1)<br>of [edge] <b>dislocations</b> (1)<br>Don't accept any reference to planes of atoms slipping over each other – needs movement of dislocation | 3               |     |     | 3     |       |      |
|          | (b) | (i)   | $\epsilon = 7.5 \times 10^{-4}$ <b>or</b> $7.5 \times 10^{-2}\%$ - no unit penalty                                                                                                                                                                                                                                                                                                      |                 | 1   |     | 1     | 1     |      |
|          |     | (ii)  | $\sigma = \epsilon \times E$<br>$= 200 \times 10^9 \times 7.5 \times 10^{-4}$ (1) (substitution) <b>(ecf on <math>\epsilon</math>)</b><br>$= 1.5 \times 10^8$ [N m <sup>-2</sup> or Pa] (1)                                                                                                                                                                                             | 1               | 1   |     | 2     | 2     |      |
|          |     | (iii) | Force = $\sigma \times \text{CSA}$<br>$= 1.5 \times 10^8 \times \pi (0.3 \times 10^{-3})^2$ (1) (conversion and substitution) <b>(ecf on <math>\sigma</math>)</b><br>$= 42[.4 \text{ N}]$ (1)<br><b>N.B. no marks for using diameter but ecf for powers of 10 error in radius e.g. using <math>0.3^2</math></b>                                                                         |                 | 2   |     | 2     | 2     |      |
|          |     | (iv)  | $W = \frac{1}{2} F \Delta l$<br>$= \frac{1}{2} \times 42.4 \times 1.8 \times 10^{-3}$ (1) <b>(ecf on force)</b><br>$= 0.038$ [J] or $0.04$ [J] (1)<br><b>Alternative:</b><br>Use of $\frac{1}{2} kx^2$ substitution (1) and answer (1)                                                                                                                                                  | 1               | 1   |     | 2     | 2     |      |